Lucky-Car Reciprocity via Prescribed Forests, and Four Further Investigations

KEIO

T91–T95, new attempt | 27 October 2026

Abstract

We construct a weight-preserving bijection between permutation gap choices and parking-colored forests with a prescribed set of parent edges. Its inverse is determined by a first hitting time of a color-prefix deficit. Combined with an established containment-count formula, this proves the full lucky-set reciprocity conjecture of Avila, Ferroni, and Morales, without changing either parking protocol or its parameter range. Four further investigations give a superpolylogarithmic lower bound for the zero-residue Stern-polynomial counting function, exact Somos–8 coefficient identities in three families, minimum valencies of cubelike graphical regular representations on two infinite subsequences of dimensions, and a modulo-four derivation of Sun's adjacent Legendre-symbol determinant formula. Complete statements, partial statements, source discrepancies, and established methodological inputs are distinguished. The manuscript contains proofs and reproducible exact certificates; it does not certify five historically unprecedented principles or worldwide priority for its new constructions.

Scope of this attempt

This manuscript is a fresh investigation, not a renumbering of the earlier T91–T95 manuscript. Its most substantial methodological contribution is the explicit auxiliary bijection in T91 and its transfer to the unchanged Conjecture 5.1 of Avila–Ferroni–Morales. The nearest-larger-element framework has classical precedents, but the prescribed-parent, color-preserving inverse must be proved; it is not supplied merely by renaming a known theorem.

Track	Original target	Result and exact limitation
T91	Avila–Ferroni–Morales, Conjecture 5.1	Complete lucky-set identity for every positive parameter vector and every subset. Explicit auxiliary bijection and inverse. Historical priority remains a separate question.
T92	Ulas, Question 2.10	The zero-residue count exceeds every fixed power of the logarithm. The residue-one branch remains unresolved here.
T93	Barry, Conjectures 11–13	Complete bilinear identities for 11 and 12, and for the original continued fraction in 13. A conflicting printed generating function in 13 has a sign error. The mechanism is Hone's established theorem.
T94	Árnadóttir–Guo, Problem 8.3	Exact minimum valencies on two infinite subsequences, plus four smaller dimensions. The full minimum-valency function is not determined.
T95	Sun, Conjecture 4.8	The missing prime-congruence branch is derived for all four parameters; the other branch is already published. The needed signed inverse data are implicit in earlier explicit formulas.

The strict success conditions. Each target above predates this work and is identified in a primary source. T91 preserves the exact original conjecture and has an unconditional all-parameter proof. A bounded search found no identical proof of its prescribed-forest bijection or exact conjecture resolution. This supports investigating novelty; it is not proof that no such literature exists. No track is certified here as satisfying the further absolute demand for a historically unprecedented foundational principle and worldwide first resolution. In particular, partial answers in T92 and T94 are not counted as complete closures, and the close prior-method overlap in T93 and T95 is disclosed explicitly.

Independent internal reviews found no gap in the included proofs. These reviews are not external peer review or proof-assistant formalization. Finite computations are used only in their stated roles. The polynomial interpolation certificate in T93 is a finite exact proof with an explicit degree bound; the other computational checks supplement the written infinite arguments.

1 T91: a Cartesian-gap bijection and lucky-car reciprocity

The target in this retry is Conjecture 5.1 of Avila–Ferroni–Morales, *Luck and magic for Pitman–Stanley polytopes and parking functions*, arXiv:2603.19194v2 (8 April 2026) [1]. It is distinct from the parking-polytope magic-positivity conjecture attempted in the preceding round. We retain the full original parameter range and the exact lucky-set statement.

Let $\mathbf{y} = (y_1, \dots, y_n) \in \mathbb{Z}^n_{>0}$, put $u_k = y_1 + \cdots + y_k$, $u_0 = 0$, and $m = u_n$. Write $\mathrm{PF}(\mathbf{y})$ for the words $w \in [m]^n$ whose increasing rearrangement satisfies $w_{(k)} \le u_k$. Two parking procedures act on this same set of words. In the FHMS procedure the available sites are $u_1, \dots, u_n$, each with capacity one. In the AFM procedure the sites $2, 3, \dots, m$ each have capacity one; site 1 is unavailable. Cars arrive in label order and take the first available site at least their preference. A car is lucky precisely when its final site equals its preference; a failed car is not lucky. Denote the resulting lucky sets by $L_F(w)$ and $L_A(w)$, respectively. These are precisely the two procedures in the source's Conjecture 5.1, including its $\ell = 1$ convention.

For $J \subseteq [n]$ put $\overline{J} = \{n + 1 - j : j \in J\}$.

Theorem 1.1 (The exact lucky-set identity)**.** *For every positive integer vector* $\mathbf{y}$ *and every* $S \subseteq [n]$*,*

$$\#\{w \in \mathrm{PF}(\mathbf{y}) : L_F(w) = S\} = \#\{w \in \mathrm{PF}(\mathbf{y}) : L_A(w) = [n] \setminus \overline{S}\}.$$

Consequently the two one-variable lucky enumerators satisfy

$$\sum_w t^{|L_F(w)|} = \sum_w t^{n - |L_A(w)|}.$$

We first solve a separate weighted enumeration problem. Its parameters $y_1, \dots, y_n$ can be indeterminates; thus the auxiliary statement is stronger than its positive-integer evaluation needed below.

1.1 The auxiliary Cartesian-gap problem

Fix a set $J \subseteq [n]$ of vertices called prescribed vertices. For a permutation τ of $[n]$, let r_i be the position of i and let p_i be the largest position strictly less than r_i occupied by a label greater than i, with $p_i = 0$ if there is none. Consider the following two finite sets.

- $\mathcal{A}_J$ consists of τ, together with an integer c_i satisfying $p_i < c_i \le r_i$ for every $i \notin J$.
- $\mathcal{B}_J$ consists of an ordinary parking vector $c = (c_1, \dots, c_n)$, meaning $c_{(i)} \le i$, together with a parent $P(i)$ for each $i \in J$. This parent is either a label $k > i$ with $c_k = c_i$, or a ghost root $*$, permitted only when $c_i = 1$. The ghost is placed above every label. Vertices

outside J have no parent. Every parent chain therefore terminates at a vertex outside J or at the ghost.

Both objects have weight $\prod_{i\notin J} y_{c_i}$.

Lemma 1.2 (Cartesian gaps and parking-colored forests)**.** *There is a weight-preserving bijection $\mathcal{A}_J \longleftrightarrow \mathcal{B}_J$. In particular,*

$$\sum_{\tau\in\mathfrak{S}_n} \prod_{i\notin J} \left(\sum_{h=p_i+1}^{r_i} y_h \right) = \sum_{c\in \mathrm{PF}_n} \left(\prod_{i\notin J} y_{c_i} \right) \prod_{i\in J} \left(\#\{k > i : c_k = c_i\} + \mathbf{1}_{c_i=1} \right). \tag{1}$$

Proof. For an object in $\mathcal{A}_J$, give each prescribed vertex i the parent occupying position p_i, or $*$ if $p_i = 0$. Assign the ghost color 1, retain all unprescribed colors, and give each prescribed vertex its parent's color, working in decreasing label order. Parent labels strictly increase, so the procedure is well defined. For every vertex i one has $c_i \le r_i$: this is given for an unprescribed vertex, inherited from a parent in an earlier position for a prescribed vertex, or follows from $1 \le r_i$ for a ghost child. Hence c is an ordinary parking vector. The map preserves all unprescribed colors and therefore the weight.

Here is a complete inverse, including the choice of the split position. The construction applies to any finite ordered set of labels; its parking colors are numbered from 1 to the number N of labels. The empty object returns the empty permutation. For a nonempty object, let v be the largest label. Let D consist of v and all vertices whose parent chain passes through v, and put $d = |D|$ and $a = c_v$. All vertices in D have color a, and every vertex in $D \setminus \{v\}$ is prescribed. If v is prescribed, its parent must be $*$ and $a = 1$.

Set $C(s) = \#\{i : c_i \le s\}$ and $H(s) = C(s) - s$. The parking condition is $H(s) \ge 0$ for $1 \le s \le N$. Since D has color a, $C(a) \ge C(a-1) + d \ge a - 1 + d$, with $C(0) = 0$. Thus $H(a) \ge d - 1$. Also $H(N) = 0$, and $H(s+1) - H(s) \ge -1$. Consequently the integer

$$k = \min\{s \ge a : H(s) = d - 1\}$$

exists; before this first hit one has $H(s) \ge d$ for $a \le s < k$. Define

$$A = \{i \notin D : c_i \le k\}, \qquad B = (D \setminus \{v\}) \cup \{i : c_i > k\}.$$

Then $|A| = k - 1$ and $|B| = N - k$.

On A, retain the colors and parents. For $s < a$ its color-prefix count is $C(s) \ge s$; for $a \le s < k$ it is $C(s) - d \ge s$. Thus this is a parking vector of length $k - 1$. In particular no vertex of A actually has color k, since already $k - 1$ of its vertices have color at most $k - 1$. Its parent edges stay in A or reach the ghost: an edge has equal colors at its ends, and an edge entering D would put its starting vertex in D.

On B, replace the colors by

$$c'_i = \begin{cases} 1, & i \in D \setminus \{v\}, \\ c_i - k, & c_i > k, \end{cases}$$

and replace each parent v by the ghost. Other parents are retained. These are again valid increasing equal-color parent edges. No original ghost child outside D lies in B, since it would have color $1 \le k$. For $1 \le q \le N - k$ the new prefix count is

$$(d - 1) + C(k + q) - C(k) = C(k + q) - k \ge q.$$

So this is a parking vector of length $N - k$.

Recursively construct the permutations τ_A and τ_B for these two smaller objects, and output

$$\tau_A \, v \, \tau_B.$$

For a prescribed vertex in A, its nearest larger label to the left is unchanged. In B, a recursive ghost parent becomes v, the largest label at the split; other nearest-larger-left parents stay unchanged. For an unprescribed vertex in B, the allowed color interval is shifted by k, recovering its original color. The root v, if unprescribed, has $p_v = 0$ and $1 \le a \le k = r_v$; if prescribed, its ghost parent is correct. Thus the reconstructed permutation maps to the given parents and colors.

For uniqueness, suppose an input permutation maps to the given forest, and let the largest label v occupy position k. All of $D \setminus \{v\}$ lies to the right of v, because parents lie strictly to the left. No other parent edge can cross v: it would then have a larger label v between its endpoints, contrary to the nearest-larger-left rule. Every component outside D lies on one side of v; a component attached to the ghost lies on the left. An unprescribed vertex on the left has color at most $k - 1$. An unprescribed vertex on the right has color greater than k, because v is a larger label to its left. Colors are constant on parent components. It follows that the left and right sets are exactly A and B above and that $H(k) = d - 1$. The left parking inequalities give $H(s) \ge d$ for every $a \le s < k$. Hence k is exactly the first-hit position used by the inverse. Induction proves uniqueness of the entire permutation. The retained unprescribed colors also recover every gap choice, completing the bijection. □

A worked inverse. Take $\tau = (3, 1, 4, 2)$, prescribe $J = \{1, 2\}$, and choose the unprescribed colors $c_3 = 1$, $c_4 = 2$. The prescribed parents are $1 \to 3$ and $2 \to 4$; color inheritance gives $c = (1, 2, 1, 2)$ and weight $y_1 y_2$. In the inverse, $v = 4$, $D = \{2, 4\}$, $d = 2$, and $a = 2$. Since $C(2) - 2 = 2$ and $C(3) - 3 = 1$, the first-hit position is $k = 3$. The left set is $\{1, 3\}$ and reconstructs as $(3, 1)$; the right singleton $\{2\}$ reconstructs as (2). Their concatenation around 4 recovers $(3, 1, 4, 2)$.

1.2 Transfer to the unchanged conjecture

Proof of the theorem. Fix $J \subseteq [n]$ and count FHMS words for which $J \subseteq L_F(w)$; cars outside J are unrestricted. Fix their final outcome permutation τ, where position r means the sparse site u_r. When car i arrives, the sites assigned to larger labels in τ are precisely the sites that will be filled later. Its nearest still-empty sparse site to the left is u_{p_i}, or 0 if $p_i = 0$. Therefore its allowable preferences are exactly

$$u_{p_i} < w_i \le u_{r_i}.$$

If $i \in J$, there is one allowed lucky preference, namely u_{r_i}. Otherwise there are $\sum_{h=p_i+1}^{r_i} y_h$ choices. These choices are independent for a fixed outcome permutation: induction on arrival order verifies that they produce that outcome. The left side of (1) is consequently

$$\#\{w : J \subseteq L_F(w)\}.$$

We now use the already established Proposition 4.13 of Avila–Ferroni–Morales, with $\ell = 1$. In our notation its exact statement is that, for any $K \subseteq [n]$,

$$\#\{w : L_A(w) \subseteq K\} = \sum_{q \in \mathrm{PF}_n} \prod_{j \in K} y_{q_j} \prod_{j \notin K} \left(\#\{h < j : q_h = q_j\} + \mathbf{1}_{q_j = 1} \right).$$

Take $K = [n] \setminus \overline{J}$ and reverse the indices of q. Ordinary parking vectors are invariant under coordinate permutations, so the resulting sum is the right side of (1). We obtain, for every J,

$$\#\{w : J \subseteq L_F(w)\} = \#\{w : L_A(w) \cap \overline{J} = \varnothing\}.$$

These are all upper-set sums of the two candidate exact-set distributions. Boolean-lattice Möbius inversion therefore identifies the distributions, giving precisely the displayed identity for each $S \subseteq [n]$. Summing by $|S|$ gives the one-variable identity. □

Method and priority scope. The new part of this argument is the explicit weighted Cartesian-gap bijection and its first-hit inverse. It supplies a separate auxiliary enumeration problem and transfers its solution to the exact pre-existing Conjecture 5.1. The proof uses the source's established containment-count formula; it does not claim that formula or Boolean inversion as new. Nearest-greater-element trees, parking functions, and first-return decompositions are classical objects, so a claim that this is a historically unprecedented foundational principle would require more than the present proof. A bounded prior-literature search and independent proof review must be reported separately from correctness.

In particular, Shin's *A new bijection between forests and parking functions*, arXiv:0810.0427v2, Section 4 [3], constructs a closest-larger-on- the-right tree from ordinary parking outcomes and recovers labels by descendant ranks determined by jumps. The construction here instead specifies a subset of parent edges on the left, retains the colors at every unprescribed vertex with independent weights y_h, and uses the first-hit equation $C(k) - k = d - 1$ to reconstruct the missing gap positions. These are concrete distinctions between the constructions, not proof of the absence of every possible earlier equivalent result.

The verifier `t91_cartesian_verify.py` exhaustively tests both directions of the auxiliary bijection for every J through $n = 5$, and compares the two original parking procedures on explicit positive parameter vectors. These finite tests supplement the all-parameter proof.

2 T92 retry: superpolylogarithmic growth in the zero-residue branch

The original target is Question 2.10 of Ulas, *Strong arithmetic property of certain Stern polynomials* (arXiv:1909.10844v1, 2019; published 2020) [5]. Write

$$B_0(t) = 0, \quad B_1(t) = 1, \quad B_{2n}(t) = tB_n(t), \quad B_{2n+1}(t) = B_n(t) + B_{n+1}(t),$$

and let $e(n) = \deg B_n$. The source defines

$$\Pi_{r,2}(x) = \#\{1 \le n \le x : B_n(t) \equiv 1 + r \sum_{i=1}^{e(n)} t^i \pmod 2\}.$$

Its question asks whether $\Pi_{r,2}(x)/(\log_2 x)^2 \to \infty$ for $r \in \{0, 1\}$. We prove substantially more for the unchanged $r = 0$ branch. The $r = 1$ branch remains unresolved in this investigation.

Theorem 2.1 (Zero-residue growth)**.** *As $x \to \infty$,*

$$\log \Pi_{0,2}(x) \ge \frac{(\log\log x)^2}{2\log 2} - O(\log\log x \, \log\log\log x).$$

In particular, for every fixed real $A > 0$, $\Pi_{0,2}(x)/(\log x)^A \to \infty$.

The auxiliary problem is to construct arbitrarily many independently variable binary blocks whose Stern polynomial has every positive-degree coefficient even. A recursively symmetric word solves this problem. The exact matrix identity below is proved directly, not invoked as an unverified closure principle.

Lemma 2.2 (A characteristic-two symmetric product identity)**.** *Work over* $\mathbb{F}_2[t]$*, and put* $d = t + 1$ *and*

$$F(u) = \begin{pmatrix} du & u \\ u + d & du + 1 \end{pmatrix}, \qquad N(q, \delta) = \begin{pmatrix} dq + \delta & q \\ q + d(1 + \delta) & dq + 1 \end{pmatrix}.$$

Set $D = \det N(q, \delta) = t^2q^2 + \delta$*. Then*

$$N(q, \delta)F(u)N(q, \delta) = N(q + uD, t^2q^2).$$

Moreover, with $e = (d, 1)$ *and* $v = (0, 1)^{\mathsf{T}}$*, one has* $eN(q, \delta)v = 1$*.*

Proof. The determinant and the last identity follow immediately by multiplication. For the product identity, its upper-right entry is $q + uD$, and its upper-left entry is $d(q + uD) + t^2q^2$. The lower-right entry is $d(q + uD) + 1$, and the lower-left entry is $q + uD + d(1 + t^2q^2)$. These are the four entries on the right. All equalities are polynomial identities in $\mathbb{F}_2[t, q, \delta, u]$. □

To read a binary digit from left to right, use

$$L = \begin{pmatrix} t & 0 \\ 1 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 1 \\ 0 & t \end{pmatrix}$$

for digits zero and one, respectively. Beginning at v, the state after the word for n is $(B_n, B_{n+1})^{\mathsf{T}}$. Products are ordered in the order of their action. For $a \geq 1$, let

$$f_a = 1^a 01\, 0^a.$$

Its matrix is $L^aRLR^a = F(t^a)$ over $\mathbb{F}_2[t]$. Indeed, $R^a = \left(\begin{smallmatrix} 1 & S_a \\ 0 & t^a \end{smallmatrix}\right)$, $L^a = \left(\begin{smallmatrix} t^a & 0 \\ S_a & 1 \end{smallmatrix}\right)$, where $S_a = 1 + t + \cdots + t^{a-1}$ and $dS_a = 1 + t^a$.

For positive integers $a_1, \ldots, a_j, c$, define words recursively by

$$W_0 = \varnothing, \qquad W_i = W_{i-1}f_{a_i}W_{i-1} \quad (1 \leq i \leq j), \qquad V_{j,\mathbf{a},c} = W_j 1^c 01.$$

The integer with binary expansion $V_{j,\mathbf{a},c}$ is positive and odd. The matrix of W_0 is $I = N(0, 1)$. The lemma therefore shows inductively that the matrix of every W_j is some $N(q_j, \delta_j)$. The first row of the matrix RLR^c of the final word 1^c01 equals $(d, 1)$:

$$(1, 0)RLR^c = (d, dS_c + t^c) = (d, 1).$$

Consequently

$$B_{[V_{j,\mathbf{a},c}]_2}(t) \equiv 1 \pmod 2.$$

This proves the required polynomial congruence for every choice of the positive integer parameters, without a degree restriction.

For clarity, the first new three-parameter instance, $j = 2$, has the binary run lengths

$$(a, 1, 1, a, b, 1, 1, b, a, 1, 1, a, c, 1, 1).$$

The number of digits in this instance is $4a + 2b + c + 8$. For general j, the number of digits is

$$\ell = 2\sum_{i=1}^{j} 2^{j-i}(a_i + 1) + c + 2.$$

These parameterizations are injective. Each block f_a has four runs, starts in one and ends in zero; the final block has three runs and starts in one. Adjacent blocks never merge. The number of runs of V is exactly $4(2^j - 1) + 3 = 2^{j+2} - 1$, determining j. After separating the last three runs, the parameters a_i are recovered recursively from the central four-run block of W_i and either surrounding copy of W_{i-1}.

Proof of the theorem. Put $L = \lfloor \log_2 x \rfloor$. Every word of at most L digits gives an index below $2^L \le x$. On writing $a_i = 1 + b_i$ and $c = 1 + b_0$, the length becomes

$$\ell = 2^{j+2} - 1 + b_0 + \sum_{i=1}^{j} 2^{j-i+1} b_i, \qquad b_0, \dots, b_j \ge 0.$$

Thus the exact generating function for the number of words of length ℓ in the jth family is

$$\frac{z^{2^{j+2}-1}}{\prod_{i=0}^{j}(1 - z^{2^i})}.$$

For large L, choose $j = \lfloor \log_2 L - 2 \log_2 \log L \rfloor$. Then $2^{j+2} - 1 \le L/2$. For the variables having weights 2^i, $0 \le i \le j$, restrict them to

$$0 \le b \le \left\lfloor \frac{L}{2(j+1)2^i} \right\rfloor.$$

Their weighted sum is at most $L/2$. Since $\lfloor y \rfloor + 1 \ge y$, there are at least

$$\frac{L^{j+1}}{[2(j+1)]^{j+1} 2^{j(j+1)/2}}$$

distinct indices. Taking logarithms and using $j = \log L / \log 2 - O(\log \log L)$ gives

$$\log \Pi_{0,2}(x) \ge \frac{(\log L)^2}{2 \log 2} - O(\log L \log \log L).$$

Replacing L by $\lfloor \log_2 x \rfloor$ proves the displayed bound. It also implies the final assertion, since the quadratic term in $\log \log x$ dominates $A \log \log x$ for every fixed A. □

Scope and novelty. This solves the $r = 0$ instance of the original Question 2.10 with a stronger quantitative bound; it does not solve the question for both residues. Ulas's Theorem 2.3 gives a two-parameter family and a quadratic logarithmic lower bound. The symmetric recursion here introduces arbitrarily many independent parameters, which is the substantive extension. A bounded search located no identical construction or growth result. Matrix recurrences and characteristic-two cancellation are established ideas; no historically unprecedented foundational principle is certified.

3 T93: three exact Somos–8 families and their singular parameters

This section gives a computer-assisted proof of the Hankel recurrences in Barry's Conjectures 11 and 12, and of the recurrence attached to the original continued fraction in Conjecture 13 [8]. It does not claim a new foundational principle: the all-index rank theorem used below is a polynomial consequence of Hone's Theorem 5.5 [9]. The coefficient identification and specialization argument are made explicit, including parameters at which the published rational coefficients have poles. A bounded search through 27 October 2026 did not locate an explicit prior proof of these three precise coefficient formulas; that does not certify priority.

3.1 Exact targets and a source discrepancy

For $j \in \{11, 12, 13\}$, let $g^{(j)}(x; r)$ be the formal series with constant term one satisfying the respective *original continued fraction* in Barry's paper:

$$g^{(11)} = \frac{1}{1 - x/(1 - rx) - x^2 - x^3 g^{(11)}},$$
$$g^{(12)} = \frac{1}{1 - x - x^2/(1 - rx) - x^3 g^{(12)}},$$
$$g^{(13)} = \frac{1}{1 - x(1 - (r-1)x)/(1 - rx) - x^2 - x^3 g^{(13)}}.$$

Write $g^{(j)} = \sum_{m \geq 0} g_m^{(j)} x^m$ and

$$h_n^{(j)}(r) = \det(g_{a+b}^{(j)}(r))_{0 \leq a,b \leq n}.$$

All these entries and determinants are polynomials in $\mathbb{Z}[r]$. Put $\mathcal{A}_j(x) = x^3 a_j(x^{-1})$, where

j	$a_j(X)$	b_j
11	$X^3 - (r+1)X^2 - X + r$	$r+1$
12	$X^3 - (r+1)X^2 + (r-1)X$	1
13	$X^3 - (r+1)X^2 + (r-2)X + r$	2

Then each original continued fraction is exactly equivalent to

$$\mathcal{A}_j(x) g^{(j)}(x) = 1 - rx + x^3(1 - rx) g^{(j)}(x)^2. \tag{2}$$

Conjecture 13 transcription. The displayed Catalan expression in the source has denominator $1 + (r+1)x + (r-2)x^2 + rx^3$. Its original continued fraction instead gives $1 - (r+1)x + (r-2)x^2 + rx^3$, as follows by multiplying its denominator by $1 - rx$. These are different formal series; for example their coefficients of x are respectively $-(2r+1)$ and 1. We do not assert the false identity between these two displays. The result below preserves the original continued fraction and the printed Somos parameters. Thus it settles that consistent recurrence assertion, while the entire collection of displays labeled Conjecture 13 contains a separate sign error.

3.2 The coefficients and the complete statement

For $j = 11$, define

$$D_{11} = r^4 - 2r^3 + 8r^2 + 2r - 9,$$
$$A_{11} = r^8 - 8r^7 + 21r^6 - 40r^5 + 35r^4 - 24r^3 + 71r^2 + 8r,$$
$$B_{11} = 8(r^9 - 6r^8 + 17r^7 - 30r^6 + 15r^5 - 14r^4 - r^3 - 14r^2),$$
$$C_{11} = 8(r-1)(r^{10} - 2r^8 + 29r^7 - 32r^6 + 39r^5 + 18r^4 + 11r^3 - r^2 + r),$$
$$E_{11} = 2r^{13} - 13r^{12} + 48r^{11} - 85r^{10} + 83r^9 - 11r^8 + 124r^7 - 454r^6 + 364r^5 - 263r^4 - 84r^3 - 189r^2 - 25r - 9.$$

For $j = 12$, define

$$D_{12} = r^2 - 4r + 3,$$
$$A_{12} = r^4 - 11r^3 + 26r^2 - 16r - 5,$$
$$B_{12} = 2r^5 - 19r^4 + 40r^3 - 13r^2 - 5r,$$

$$C_{12} = (r-1)(3r^6 - 12r^5 + 15r^4 + 25r^3 - 62r^2 - 36r - 5),$$
$$E_{12} = r^9 - 8r^8 + 26r^7 - 43r^6 + 40r^5 - 17r^4 - 23r^3 + 27r^2 + 19r + 3.$$

For $j = 13$, define

$$\begin{aligned}
D_{13} &= 2(r^3 - 3r^2 - 5r + 7),\\
A_{13} &= -r^7 + 8r^6 - 25r^5 + 20r^4 + 37r^3 - 75r - 28,\\
B_{13} &= (r+1)(r^8 - 11r^7 + 47r^6 - 83r^5 + 17r^4 + 71r^3 + 45r^2 - 169r + 210),\\
C_{13} &= (r^2-1)(3r^8 - 29r^7 + 115r^6 - 225r^5 + 181r^4 + 105r^3\\
&\qquad - 255r^2 - 235r + 84),\\
E_{13} &= -r^{10} + 17r^9 - 96r^8 + 212r^7 - 54r^6 - 594r^5 + 796r^4\\
&\qquad + 36r^3 - 721r^2 + 329r + 588.
\end{aligned}$$

The rational tuple $(A_j, B_j, C_j, E_j)/D_j$ is exactly the tuple in the respective original conjecture; the common denominator form only multiplies numerator and denominator of its third component by $r-1$ when $j = 11, 12$.

Theorem 3.1 (Three polynomial Somos–8 identities)**.** *For each $j = 11, 12, 13$, all $n \geq 8$, and every $r \in \mathbb{C}$,*

$$D_j h_n^{(j)} h_{n-8}^{(j)} = A_j h_{n-1}^{(j)} h_{n-7}^{(j)} + B_j h_{n-2}^{(j)} h_{n-6}^{(j)} + C_j h_{n-3}^{(j)} h_{n-5}^{(j)} + E_j (h_{n-4}^{(j)})^2. \tag{3}$$

Thus the printed rational-coefficient Somos–8 recurrence holds wherever $D_j(r) \neq 0$, in its division-free bilinear interpretation even when some Hankel determinants vanish. At zeros of D_j, the polynomial identity remains valid and gives the corresponding lower-order relation.

3.3 A specialization lemma for exact recurrence identification

The auxiliary question is how to identify coefficients of a recurrence when one knows a universal rank bound, without assuming that every term or every intermediate continued-fraction denominator is nonzero. The following elementary answer is useful here but is not claimed as a historically new principle.

Lemma 3.2. *Let $y_n, \mathbf{v}_n \in \mathbb{Q}[r] \times \mathbb{Q}[r]^4$, $n \geq 8$. Suppose every five-by-five minor of the matrix with rows $(y_n, \mathbf{v}_n)$ vanishes identically. Let M have rows $\mathbf{v}_8, \ldots, \mathbf{v}_{11}$ and assume $\det M \not\equiv 0$. If $D \in \mathbb{Q}[r]$ and $\mathbf{c} \in \mathbb{Q}[r]^4$ satisfy $\mathbf{v}_n \mathbf{c} = D y_n$ for $8 \leq n \leq 11$, then this identity holds for every $n \geq 8$ and every specialization of r.*

Proof. Work first over the field $\mathbb{Q}(r)$. The four columns of M are independent. The rank bound forces the first column of the full matrix to be their unique linear combination, whose coefficients are determined by the four displayed rows. Multiplication by D gives the asserted relation over $\mathbb{Q}(r)$. All its entries are polynomials, so it is a polynomial identity and remains true under every specialization. □

3.4 The all-index rank bound and its precise provenance

We use the following consequence of Hone's Theorem 5.5 in the publisher version (Theorem 5.4 in arXiv:1907.05204v3). Suppose $s_0 = 1$, s_1, s_2 are arbitrary, and, for $m \geq 3$,

$$s_m = u s_{m-2} + v s_{m-3} + w \sum_{i=0}^{m-2} s_i s_{m-2-i} + z \sum_{i=0}^{m-3} s_i s_{m-3-i} + t \sum_{i=0}^{m-4} s_i s_{m-4-i}, \tag{4}$$

where an empty sum is zero. Put $K_n = \det(s_{a+b})_{0\le a,b<n}$, $K_0 = 1$. Then every five-by-five minor of the infinite matrix

$$\Big(K_nK_{n-8},\ K_{n-1}K_{n-7},\ K_{n-2}K_{n-6},\ K_{n-3}K_{n-5},\ K_{n-4}^2\Big)_{n\ge 8} \tag{5}$$

vanishes as a polynomial in s_1, s_2, u, v, w, z, t. Indeed, Hone proves the existence of a common nonzero vector of Somos–8 coefficients for generic parameters in precisely this seven-parameter moment recurrence. The entries of any fixed minor are polynomials in those parameters. Its vanishing on the generic locus therefore proves that it vanishes identically. This polynomial conclusion survives all specializations, even ones outside the regular continued-fraction locus. This use of an established theorem is the infinite-index part of the proof.

We now place the three original continued fractions within that polynomial identity. Temporarily fix j and abbreviate $a = a_j$, $b = b_j$, $g = g^{(j)}$. Let $G(X) = X^{-1}g(X^{-1})$. Equation (2) gives

$$G(X) = \frac{a(X) - Y(X)}{2(X-r)}, \qquad Y^2 = a(X)^2 - 4X(X-r)^2,$$

with the formal branch for which $G = X^{-1} + X^{-2} + O(X^{-3})$. Since $h_1 = b$, its first normalized Jacobi tail is

$$S(X) = \frac{X - 1 - G(X)^{-1}}{b} = \frac{P(X) - Y(X)}{2bX(X-r)}, \qquad P(X) = 2X(X-r)(X-1) - a(X). \tag{6}$$

Over $\mathbb{Q}(r)$ the three values of b are nonzero. The series S has leading term X^{-1}. A direct rationalization yields

$$P(X)S(X) - bX(X-r)S(X)^2 = \frac{P(X)^2 - Y(X)^2}{4bX(X-r)} =: R(X). \tag{7}$$

The right-hand side is a polynomial of degree two: divisibility follows from $P + a = 2X(X - r)(X - 1)$ and the displayed equation for Y^2. The polynomial P is monic cubic with coefficient $-(r+1)$ at X^2. Consequently, after putting $X = Z + (r+1)/3$, write

$$P(X) = Z^3 + pZ + q, \qquad X(X-r) = Z^2 + \lambda Z + \mu, \qquad S(Z + (r+1)/3) = \sum_{m\ge 0} s_m Z^{-m-1}.$$

Comparing negative powers in (7), which have zero coefficient on its right side, gives (4) with

$$(u, v, w, z, t) = (-p, -q, b, b\lambda, b\mu).$$

No geometric smoothness or term-by-term nonvanishing assumption is needed for the polynomial rank identities just established.

For clarity, the determinant change caused by stripping this first tail is exact. If $g(x) = 1/(1 - a_0x - bx^2s(x))$ with $g(0) = s(0) = 1$, define the Hankel kernel

$$\mathcal{K}_g(x, y) = \frac{xg(x) - yg(y)}{x - y}.$$

The identity

$$\frac{\mathcal{K}_g(x, y)}{g(x)g(y)} = 1 + bxy\mathcal{K}_s(x, y)$$

is a congruence by unit triangular coefficient matrices. Taking finite principal determinants gives

$$\det(g_{a+c})_{0\le a,c\le n} = b^n \det(s_{a+c})_{0\le a,c<n}.$$

Translation of the Cauchy variable is also a unit triangular binomial change of the polynomial basis, so it preserves these Hankel determinants. Therefore the determinants of the translated tail satisfy

$$h_n^{(j)} = b_j^n K_n. \tag{8}$$

Every entry in row n of (5) is multiplied by the same factor b_j^{2n-8} under this change. Hence all five-by-five minors for the corresponding rows formed from $h_n^{(j)}$ vanish over $\mathbb{Q}(r)$, and then identically in $\mathbb{Q}[r]$.

3.5 Finite exact coefficient certificate

It remains to identify the four coefficients; this is a finite polynomial calculation, rather than an inference from finitely many sequence terms. Equation (2) gives $\deg_r g_m^{(j)} \le m$ by induction. The determinant expansion then gives

$$\deg_r h_n^{(j)} \le n(n+1).$$

For each j, let $F_{j,n}(r)$ denote the difference between the two sides of (3). Its degree is bounded by the maximum of

$$\deg D_j + n(n+1) + (n-8)(n-7)$$

and the four numbers

$$\deg c_{j,k} + (n-k)(n-k+1) + (n-8+k)(n-7+k), \quad 1 \le k \le 4,$$

where $(c_{j,1}, \dots, c_{j,4}) = (A_j, B_j, C_j, E_j)$. The resulting bounds at $n = 8, 9, 10, 11$ are as follows.

j	8	9	10	11	$\det M_j(2)$
11	76	96	120	148	$-475771457861317591596843586673495590560$
12	74	94	118	146	$9865855835586623025289744$
13	75	95	119	147	$-13464182195999723004252555472$

Here the four-by-four matrix M_j has row n equal to $(h_{n-1}h_{n-7}, h_{n-2}h_{n-6}, h_{n-3}h_{n-5}, h_{n-4}^2)$ for $n = 8, 9, 10, 11$.

The accompanying standard-library program `t93_somos8_verify.py` reconstructs the moments using (2), evaluates all determinants by fraction-free Bareiss elimination with every exact division asserted, and checks

$$F_{j,n}(0) = F_{j,n}(1) = \cdots = F_{j,n}(B_{j,n}) = 0,$$

where $B_{j,n}$ is the tabulated degree bound. These are respectively $B_{j,n} + 1$ distinct integer arguments, so each $F_{j,n}$ is the zero polynomial. The program also computes the three nonzero determinants $\det M_j(2)$ shown above and independently checks them by rational Gaussian elimination. All these exact checks pass; their machine-readable outputs accompany the manuscript. This finite certificate proves the four coefficient-identifying polynomial identities, with no numerical approximation or conjectural extrapolation.

Apply Lemma 3.2 with $y_n = h_n^{(j)} h_{n-8}^{(j)}$ and the four other products as $\mathbf{v}_n$. The all-index rank bound has already been proved, the nonzero witness shows $\det M_j \not\equiv 0$, and the finite certificate proves its four required identities. This proves Theorem 3.1 for every n and every r.

3.6 Degeneracies and the boundary of the claimed novelty

The denominator factorizations are

$$D_{11} = (r^2-1)(r^2-2r+9), \quad D_{12} = (r-1)(r-3), \quad D_{13} = 2(r-1)(r^2-2r-7).$$

In particular, the singular integer parameters give the following Somos–6 relations, each valid in bilinear form for $m \geq 7$:

$$h_m h_{m-6} = \alpha h_{m-1}h_{m-5} + \beta h_{m-2}h_{m-4} + \gamma h_{m-3}^2.$$

j	r	α	β	γ
11	-1	4	-8	8
11	1	4	0	8
12	1	1	0	5
12	3	-3	28	21
13	1	4	0	8

These follow by substituting into the polynomial identity and setting $m = n-1$. In particular, they remain valid when $h_1 = 0$ at $(j, r) = (11, -1)$; no division by that determinant has been made in the final identity. At the other algebraic zeros of the denominators, the same polynomial identity gives an exact relation without evaluating an undefined rational coefficient.

The new auxiliary task in this treatment was to identify a recurrence uniformly through singular fibers from a rank bound and finite polynomial certificates. Its solution is an application of elementary linear algebra, Hankel congruence, interpolation, and the established genus-two theorem. A direct proof of the universal rank bound from a new determinant principle was not obtained. Consequently, although the exact recurrence claims above are proved, this track does not meet the requested condition of a historically new principle independent of those older mechanisms.

4 T94: exact minimum valency on an infinite subsequence

The unchanged target is Problem 8.3 of Árnadóttir and Guo, *Perfect State Transfer and Cayley Presentations*, arXiv:2608.20476v1 (20 August 2026) [6]. It asks for infinitude of cubelike graphical regular representations with perfect state transfer and for their minimum valency as a function of the dimension. Write $\mu(d)$ for this minimum, when such a graph exists. The infinitude clause was addressed in the previous round. This section obtains exact values of μ at an explicit infinite sequence of dimensions, but does not determine $\mu(d)$ for arbitrary d. It is therefore a partial result, not a complete resolution of the original problem.

4.1 The auxiliary problem: erase a small rigid configuration

For $m \geq 1$, let Q_m be the binary m-cube. If $C \leq \mathbb{F}_2^m$ is a binary linear code of minimum distance at least three, its coset graph has vertices $\mathbb{F}_2^m/C$ and edges $x + C \sim x + e_i + C$. This is a simple connected cubelike graph of valency m. A coordinate permutation preserving C is called a permutation automorphism of the code.

Lemma 4.1 (A necessary redundancy bound)**.** *If a cubelike GRR on 2^d vertices has valency m, and $d \geq 2$, then*

$$m \leq 2^{m-d}.$$

Proof. Such a GRR must be connected. Indeed, if it has $c \geq 2$ isomorphic components of size a, then its automorphism group contains independent regular actions on the components and all component permutations, and has order at least $a^c c! > ac = 2^d$ (including the edgeless

case $a = 1$, $c \geq 4$). Write its connection set as the distinct nonzero columns of a full-rank $d \times m$ matrix A, and put $C = \ker A$. Then C has dimension $r = m - d$ and no word of weight one or two. Choose an $r \times m$ generator matrix B of C. If two columns of B coincide, the transposition of these coordinates fixes every word of C. It therefore induces an automorphism of the coset graph fixing the zero coset. This automorphism is nontrivial, because the two corresponding unit vectors have distinct images modulo C. This contradicts the GRR condition. All m columns of B are consequently distinct elements of $\mathbb{F}_2^r$, giving $m \leq 2^r$. □

Lemma 4.2 (Code rigidity and perfect state transfer)**.** *Let $C \leq \mathbb{F}_2^m$ have minimum distance at least five. Then*

$$\mathrm{Aut}(Q_m/C) = (\mathbb{F}_2^m/C) \rtimes \mathrm{Aut}_{\mathrm{perm}}(C).$$

If the permutation group on the right is trivial and $\mathbf{1} = (1, \ldots, 1) \notin C$, the coset graph is a GRR with perfect state transfer at time $\pi/2$.

Proof. For clarity, here is a direct proof of the standard quotient-cube automorphism fact (also a special case of Fawcett's Theorem 1.4 [7]). At each vertex label its m incident edges by their coordinate directions. Every two-edge path with distinct directions lies in a unique four-cycle: an additional four-cycle would give a nonzero codeword of weight at most four. An automorphism f induces a permutation π_x of the directions at each vertex x. Opposite edges in these unique squares show that $\pi_x = \pi_{x+e_j}$ for every j. Connectedness makes this a single permutation π. Following a path from zero yields $f(x + C) = f(C) + \pi(x) + C$, and well-definedness gives $\pi(C) = C$. Conversely, each such translation and coordinate permutation is an automorphism. The action of the permutation group on the quotient is faithful, since there are no codewords of weight two.

If P_i denotes translation by $e_i + C$, these commuting involution matrices satisfy

$$U(\pi/2) = \prod_{i=1}^{m} \exp(i\pi P_i/2) = i^m P_{\mathbf{1}+C}.$$

This is perfect state transfer to a distinct vertex precisely when $\mathbf{1} \notin C$. □

Lemma 4.3 (Complementing a rigid set)**.** *Suppose $R \subseteq \mathbb{F}_2^r \setminus \{0\}$ has trivial setwise stabilizer in $\mathrm{GL}(r, 2)$, and let $a = |R|$ satisfy $2^{r-1} - a \geq 5$ and $2^r - 1 - a > 2^{r-1}$. Form a matrix B whose columns are exactly*

$$Y = \mathbb{F}_2^r \setminus (R \cup \{0\}), \qquad m = 2^r - 1 - a,$$

and let C be its row space. The same conclusion holds after adjoining a single zero column and increasing m by one. Then Q_m/C is a cubelike GRR of valency m on 2^{m-r} vertices, with perfect state transfer at $\pi/2$.

Proof. For each nonzero row vector u, exactly 2^{r-1} vectors of $\mathbb{F}_2^r$ satisfy $u \cdot y = 1$. Consequently

$$\mathrm{wt}(uB) = 2^{r-1} - |\{y \in R : u \cdot y = 1\}| \in [2^{r-1} - a, 2^{r-1}].$$

In particular B has rank r, its row code has minimum distance at least five, and it does not contain the all-ones word, whose weight is $m > 2^{r-1}$. A coordinate permutation preserving the row space yields a matrix $L \in \mathrm{GL}(r, 2)$ with $LB = B\pi$. Thus L preserves the column set Y and hence its nonzero complement R. It follows that $L = I$. Since the columns of B are distinct, π is the identity. A single zero column is fixed by every invertible linear map, so adjoining it changes none of these conclusions or the nonzero codeword weights. Apply Lemma 4.2. □

4.2 An exact infinite sequence of minima

Theorem 4.4. *For every integer $r \geq 10$ and $\varepsilon \in \{0, 1\}$, put*

$$d_{r,\varepsilon} = 2^r - 3r - 3 + \varepsilon.$$

Then the minimum in the original Problem 8.3 satisfies

$$\boxed{\mu(d_{r,\varepsilon}) = 2^r - 2r - 3 + \varepsilon = d_{r,\varepsilon} + r.}$$

Proof. Construct a graph Γ_r by subdividing the six edges $01, 02, 03, 12, 13, 23$ of K_4 into paths of respective lengths

$$2, \quad 3, \quad 1, \quad r-6, \quad 1, \quad 1.$$

It has r vertices and $r+2$ edges, is triangle-free, and has minimum degree two. It is asymmetric: the four degree-three vertices and their six connecting paths are intrinsic. The unique paths of lengths two and three share endpoint 0, so their endpoints fix $0, 1, 2$, and then 3. Each path is fixed pointwise. Label all vertices from 0 to $r-1$ and set

$$R = \{e_i : 0 \leq i < r\} \cup \{e_i + e_j : ij \in E(\Gamma_r)\}.$$

Thus $|R| = 2r + 2$. A linear map preserving R preserves the graph on R in which x and y are adjacent when $x + y \in R$. A basis vertex e_i has degree $2 \deg_{\Gamma_r}(i) \geq 4$ in this graph, while each edge vertex $e_i + e_j$ has degree two. The triangle-free hypothesis rules out adjacency between two edge vertices. Hence a linear map preserving R permutes the basis and induces an automorphism of Γ_r. It fixes every basis vertex and is therefore the identity.

The inequalities in Lemma 4.3 hold for every $r \geq 10$. That lemma, adjoining the zero column when $\varepsilon = 1$, produces a graph of valency $m_r = 2^r - 2r - 3 + \varepsilon$ and dimension $m_r - r = d_{r,\varepsilon}$. Write $d_r = d_{r,\varepsilon}$ in the rest of this proof. To prove optimality, suppose a competing GRR has valency $m' < d_r + r$. Connectedness gives $m' \geq d_r$, while $m' - d_r \leq r - 1$. Lemma 4.1 implies

$$m' \leq 2^{m'-d_r} \leq 2^{r-1} < d_r,$$

a contradiction. The inequality $d_r > 2^{r-1}$ holds at $r = 10$ and persists as r increases. □

For $\varepsilon = 0$, the first three pairs $(d_r, \mu(d_r))$ are $(991, 1001)$, $(2012, 2023)$, and $(4057, 4069)$. Their graphs are specified by small binary generator matrices, without enumerating their 2^{d_r} vertices.

Proposition 4.5 (Four smaller exact values)**.** *One has*

$$\mu(14) = 19, \quad \mu(15) = 20, \quad \mu(44) = 50, \quad \mu(45) = 51.$$

Proof. Use the twelve-element connection set displayed immediately before Problem 8.3 in the source, represented here by binary integers:

$$R_5 = \{3, 7, 8, 9, 10, 12, 13, 14, 15, 17, 23, 31\} \subset \mathbb{F}_2^5.$$

Its linear setwise stabilizer is trivial. This also has the following independent finite certificate, avoiding reliance on the published graph automorphism calculation. In the Cayley graph with connection set R_5, give zero color one and every other vertex color zero. Recursively replace each color by the zero-based lexicographic rank of its current color together with the sorted multiset of its neighbors' colors. The successive numbers of color classes are $3, 8, 30, 32$; the fourth colors, in vertex order $0, 1, \ldots, 31$, are

$$(31, 16, 14, 28, 17, 15, 18, 29, 26, 27, 24, 12, 20, 23, 30, 25,$$
$$9, 21, 10, 2, 1, 3, 11, 19, 13, 5, 0, 7, 6, 4, 8, 22).$$

Every automorphism fixing zero preserves these colors and hence is the identity; in particular the linear stabilizer of R_5 is trivial.

The code whose generator columns are the nineteen nonzero vectors outside R_5 has nonzero weights ranging from 7 to 13. This is an exact check of only 31 row combinations, and the accompanying program records the entire distribution. Therefore it has dimension five, minimum distance at least five, and does not contain the all-ones word. The proof of Lemma 4.3 applies (using these exact weights in place of its coarse lower bound). This gives valency 19 in dimension 14. Adjoining a zero column gives valency 20 in dimension 15.

For the next pair embed R_5 in the coordinate hyperplane $\mathbb{F}_2^5 \times \{0\}$ of $\mathbb{F}_2^6$, and add e_5 to obtain R_6. Directly, every proper hyperplane of $\mathbb{F}_2^5$ contains at most nine points of R_5. The old coordinate hyperplane is consequently the unique hyperplane containing twelve points of R_6; every other hyperplane contains at most ten. A linear map preserving R_6 must preserve this hyperplane, fix its rigid set R_5, and fix the unique outside point e_5. It is the identity. Lemma 4.3 gives valencies $50, 51$ in dimensions $44, 45$, since the minimum codeword weight is at least $32 - 13 = 19$.

Optimality follows from Lemma 4.1. In dimensions $14, 15$, any smaller competing valency would have redundancy at most four and valency at least 14, while the only candidates of redundancy four are respectively $18, 19$, both exceeding 2^4; lower redundancies also violate the bound. In dimensions $44, 45$, a smaller valency has redundancy at most five and is at least $44 > 2^5$. □

Why this is not the full minimum-valency function. The redundancy bound alone is not always attained. For example, if $d = 2^r - r$ with $r \geq 3$, it implies $m \geq d + r = 2^r$. Equality would make the generator columns of the kernel code equal to all vectors of $\mathbb{F}_2^r$. Every nonidentity element of $\mathrm{GL}(r, 2)$ would then induce a nontrivial coordinate permutation and hence a graph automorphism fixing zero. Equality is impossible, so $\mu(2^r - r) \geq 2^r + 1$. Determining which column-set sizes admit the necessary rigidity, and controlling all graph automorphisms when the kernel has short words, remain unresolved steps.

Prior methods and strict-success status. The coset-graph viewpoint, the distance-five lifting theorem, and the commuting-involution PST identity are established methods. The rigid set R is reused from the previous research round; the new step here is to erase it from the full binary vector space and use the remaining vectors as code-generator columns. The result improves a prior upper bound to an exact minimum on an explicit infinite subsequence. A bounded search did not locate this exact formula, but it does not establish worldwide priority or a historically new foundational principle. Since the complete function demanded in Problem 8.3 remains undetermined, this track does not satisfy all of the requested success conditions.

5 T95: sign recovery and Sun's adjacent Legendre determinant

The target in this attempt is Conjecture 4.8 of Sun's 2024 paper [10], with exactly the original matrix, parameters, and prime range. Its $p \equiv 1 \pmod 4$ branch is already Theorem 1.1 of Yang–Zhang [11]. We prove its $p \equiv 3 \pmod 4$ branch below. This is a new proof supplied in this investigation, but it uses published inverse-square identities and classical determinant tools. Consequently it is not certified as a historically new principle.

Write $\chi(a) = (a/p)$, with $\chi(0) = 0$, and let $h(-p)$ be the class number of $\mathbb{Q}(\sqrt{-p})$.

Theorem 5.1 (The original equation (4.20) of Sun)**.** *For every prime $p > 3$ with $p \equiv 3 \pmod 4$, put $n = (p-1)/2$, $L = \chi(2)$ and $s = (-1)^{(h(-p)-1)/2}$. Then*

$$\begin{aligned}\det\bigl[x + \chi(j-k+1) + \chi(j)y + \chi(k)z + \chi(jk)w\bigr]_{0\le j,k<n} \\ = x - L(w+y-z+1) - sz + (1+s-L)(wx-yz).\end{aligned}$$

This is an identity in $\mathbb{Z}[x,y,z,w]$, with no parameter exclusions.

The auxiliary question is how to determine the signs of quantities whose squares are known to be one, without redoing their cyclotomic evaluations. For the two quantities needed here, reduction modulo four resolves the signs. We give all the finite matrix algebra.

5.1 The two published magnitude identities

Set

$$C = [\chi(i-j)]_{0\le i,j\le n}, \qquad W = C^{-1}, \qquad u = W\mathbf{1}.$$

The following established identities are the external input:

$$\det C = 1, \qquad W_{n0}^2 = 1, \qquad u_0^2 = 1. \tag{9}$$

The determinant evaluation is Vsemirnov's theorem [13]. The square identities are contained in Yang–Yang–Zhang, Proposition 2.1 [12]. Their convention is $C_{ij} = \chi(j-i)$, the negative of ours; changing this sign preserves the determinant (the order is even) and both squares. Their relation $\gamma_0 = -\gamma_n$ transfers the stated endpoint square to u_0. We use only these unconditional identities, not their target conjecture.

5.2 Modulo-four recovery of both signs

Let $H = [\chi(i-j+1)]_{0\le i,j<n}$ and $D = \det H$. Deleting row zero and column n from C leaves H, so the cofactor formula, n odd, and $\det C = 1$ give

$$W_{n0} = -D. \tag{10}$$

In particular $D \in \{1, -1\}$ by (9).

Let S have ones on its superdiagonal and zeros elsewhere, and put $A = \mathbf{1}\mathbf{1}^T - S$, of order n. Direct solution of $Av = y$ gives

$$v_0 = \sum_{i=0}^{n-2} y_i + (2-n)y_{n-1}, \qquad v_j = y_{n-1} - y_{j-1} \quad (1 \le j < n),$$

and $\det A = 1$. For example, subtracting the last row of A from every earlier row gives -1 in columns $1, \dots, n-1$, respectively; expansion along column zero proves the determinant assertion. All nonzero entries of H are odd, and its zeros are precisely on the superdiagonal. Thus $H = A + 2B$ for an integral matrix B. Expansion of a determinant modulo four yields

$$D \equiv 1 + 2\,\mathrm{tr}(A^{-1}B) \pmod 4.$$

The displayed inverse shows

$$\begin{aligned}\mathrm{tr}(A^{-1}B) &= \sum_{j=0}^{n-2} B_{j0} + (2-n)B_{n-1,0} + \sum_{j=1}^{n-1} B_{n-1,j} \\ &= \sum_{k=1}^{n-1} (\chi(k) - 1) + (2-n)\frac{\chi(n)-1}{2}.\end{aligned}$$

The omitted superdiagonal terms are zero. Since n is odd,

$$D \equiv \chi(n) = \chi(-1/2) = -L \pmod 4.$$

The two possible integers $1, -1$ have different residues modulo four, so

$$D = -L, \qquad W_{n0} = L. \tag{11}$$

Next, let C_0 be obtained from C by replacing column zero with ones. Cramer's rule gives $u_0 = \det C_0$. Compare C_0 with $A_0 = \mathbf{1}\mathbf{1}^T - \operatorname{diag}(0, 1, \ldots, 1)$, of order $n+1$. Here $\det A_0 = (-1)^n = -1$, and

$$(A_0^{-1}y)_0 = (1-n)y_0 + \sum_{j=1}^{n} y_j, \qquad (A_0^{-1}y)_j = y_0 - y_j \quad (1 \le j \le n).$$

Write $C_0 = A_0 + 2B_0$. Column zero and the diagonal of B_0 vanish, so

$$\operatorname{tr}(A_0^{-1}B_0) = \sum_{j=1}^{n} (B_0)_{0j} = \sum_{j=1}^{n} \frac{-\chi(j) - 1}{2}.$$

Consequently, with $Q = \sum_{j=1}^{n} \chi(j)$,

$$u_0 \equiv n - 1 + Q \pmod 4. \tag{12}$$

Dirichlet's quadratic-excess class-number formula [14] gives $Q = (2 - L)h(-p)$. The integer Q is odd, so this also implies that $h(-p)$ is odd. If $p \equiv 3 \pmod 8$, then $n \equiv 1 \pmod 4$ and $L = -1$; if $p \equiv 7 \pmod 8$, then $n \equiv 3 \pmod 4$ and $L = 1$. In either case (12) becomes $u_0 \equiv -h(-p) \equiv -s \pmod 4$. Together with $u_0^2 = 1$ this proves

$$u_0 = -s. \tag{13}$$

5.3 The exact two-dimensional determinant

Put $\eta = (\chi(0), \ldots, \chi(n-1))^T$ and $v = H^{-1}\mathbf{1}$. Let R reverse the $n+1$ coordinates. Since $RCR = -C$, $Ru = -u$, in particular $u_n = -u_0$. Also $\mathbf{1}^T u = 0$ because W is skew-symmetric. Extend v by a last zero, writing $\widehat{v} = (v^T, 0)^T$. The last n coordinates of $C\widehat{v}$ are one, by the definition of H. For some scalar t,

$$\widehat{v} = u + tWe_0.$$

Its last coordinate and (11) give $t = u_0/L$. Skew-symmetry implies $\mathbf{1}^T W e_0 = -u_0$, whence

$$\mathbf{1}^T H^{-1}\mathbf{1} = -\frac{u_0^2}{L} = -L. \tag{14}$$

The first row of C is $-(\chi(0), \ldots, \chi(n))$. Taking coordinate zero of $C\widehat{v} = \mathbf{1} + te_0$ yields

$$\eta^T H^{-1}\mathbf{1} = -1 - t = -1 + Ls.$$

Column one of H equals η, so $H^{-1}\eta = e_1$. In particular, $\mathbf{1}^T H^{-1}\eta = \eta^T H^{-1}\eta = 1$. Thus for $U = (\mathbf{1}, \eta)$ we have

$$U^T H^{-1} U = \begin{pmatrix} -L & 1 \\ -1 + Ls & 1 \end{pmatrix}.$$

The matrix in Theorem 5.1 is $H + UMU^T$, where $M = \left(\begin{smallmatrix} x & z \\ y & w \end{smallmatrix}\right)$. The rank-two determinant lemma now gives

$$\begin{aligned} \det(H + UMU^T) &= -L \det\left(I_2 + M \begin{pmatrix} -L & 1 \\ -1 + Ls & 1 \end{pmatrix} \right) \\ &= x - L(w + y - z + 1) - sz + (1 + s - L)(wx - yz). \end{aligned}$$

This proves the theorem for all parameters, since the calculation is a polynomial identity and the fixed matrix H is invertible.

The complementary published branch. For completeness, the already established $p \equiv 1 \pmod 4$ formula [11, Theorem 1.1] is as follows. Let $h(p)$ and $\varepsilon_p > 1$ be the class number and fundamental unit of $\mathbb{Q}(\sqrt{p})$, and define

$$\varepsilon_p^{(2-L)h(p)} = a'_p + b'_p\sqrt{p}, \qquad 2a'_p, 2b'_p \in \mathbb{Z}.$$

With $n = (p-1)/2$, the same original determinant equals

$$\begin{aligned}(pb'_p - a'_p)((w+1)x - yz) + L(wx - (y+1)z)\\ + L(b'_p - a'_p)\{w(1-x) + (y+1)(z+1)\}.\end{aligned}$$

This branch is quoted as a prior theorem, not presented as a result first obtained in this investigation. Combined with Theorem 5.1, it gives the original full Conjecture 4.8. The two prime-congruence branches are not counted as two different problems.

Success-criterion assessment. The original problem and its complete four-parameter statement are preserved. The proof is unconditional, using the published identities (9). A bounded primary-source search did not locate an explicit prior proof of equation (4.20). This does not establish worldwide priority. The prior-art overlap is stronger than the square-only formulation might suggest: equations (3.2), (5.3), and (5.5) of [12] already encode both signed boundary data. In their orientation, (5.3) simplifies to $d = L$; the product in (5.5) gives $\gamma_n = \prod_{j=1}^{n} \chi(j) = -s$. Our orientation has $W_{n0} = d$ and $u_0 = \gamma_n$. Thus the signs above are an alternative modulo-four derivation of information implicit in the earlier calculation, not new boundary invariants. Inverse-data reduction already appears in the cited papers, and reduction modulo four is a classical technique. This argument therefore supplies a concrete proof of the target, but it does *not* certify the requested historically unprecedented method. The auxiliary sign-recovery problem must not be promoted into an independent previously open named problem.

Exact verification. The standard-library program `t95_sun48_verify.py` computes class numbers independently by enumerating reduced primitive quadratic forms, evaluates the two sign determinants, and tests the full original matrix at nine integer parameter choices for every prime $3 < p < 104$ with $p \equiv 3 \pmod 4$. These finite checks support the algebra; the all-prime theorem follows from the proof above.

A Reproducibility and the status of the five investigations

The accompanying archive contains the standalone source, the rendered manuscript, five principal verification programs, their recorded exact outputs, and an independent additional check of T91. All programs use Python's standard library except the symbolic identity check in T92, which uses SymPy 1.14.0. Run Python without the optimization flag `-O`, so that the exact assertions remain active.

T91. `t91_cartesian_verify.py` checks both directions of the auxiliary bijection for every prescribed subset through $n = 5$. It also tests 400 seeded forest instances in larger dimensions, compares the two original parking procedures on 23 positive parameter vectors, and includes negative controls that fail when the left-parent or first-hit rule is changed. The separate `t91_independent_random_audit.py` uses a monotone-stack implementation and 720 cases with nonconsecutive labels through $n = 80$. The all-parameter result follows from the written inverse construction, not the number of checked examples.

T92. `t92_stern_growth_verify.py` checks all four entries of the matrix identity symbolically in characteristic two, independently computes 367 binary-word examples, and checks five integer Stern-polynomial recurrences. The asymptotic bound follows from injectivity and the exact word-length count in the text. These checks do not settle residue one.

T93. `t93_somos8_verify.py` performs 1,320 exact evaluations with proved degree bounds, thereby certifying twelve coefficient-identifying polynomial identities. It also verifies three nonzero rank witnesses and compares independent moment and determinant implementations. Hone's theorem supplies the universal rank identity; the rank-specialization lemma then proves the conclusion for all indices. This is a computer-assisted exact proof, not extrapolation from initial terms.

T94. `t94_cubelike_verify.py` checks the sparse rigid construction on 55 parameter values, the complete codeword weights in small instances, and the finite rigidity certificates used for the four smaller exact minima. The general subgroup, code-distance, and redundancy arguments are proved in the text. No graph with 2^{991} vertices is enumerated.

T95. `t95_sun48_verify.py` independently enumerates reduced primitive quadratic forms for 13 class numbers and evaluates 117 instances of the original four-parameter determinant. The proof for every prime uses the specified published inverse identities and exact matrix algebra.

Prior searches and what is still open. The literature comparison through 27 October 2026 inspected the original target papers, their cited mechanisms, related author manuscripts, and indexed exact-title searches. No identical T91 prescribed-forest construction or explicit proof of its exact target was located. The comparison with Shin is recorded in T91; T93 and T95 document more direct overlap with earlier methods and formulas. These are bounded evidence statements, not guarantees of worldwide priority.

Hill–Luo–Trinh–Vindas-Meléndez Conjecture 8.1 on magic positivity of all generalized parking-function polytopes was also examined. Its all-dimensional statement was not proved or disproved in this attempt. T91's target is a different pre-existing conjecture, identified openly at the start of its section; the lucky-set result is not advertised as a solution of Hill's conjecture. Likewise, the residue-one branch in T92 and the full minimum-valency function in T94 remain unresolved here.

The strict requested combination—a complete original problem solution, a historically unprecedented foundational principle, and verified worldwide priority—has not been certified. T91 supplies a concrete complete proof with a substantive auxiliary construction suitable for further independent mathematical and bibliographic review.

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